

UST KAMENOGORSK, Kazakhstan

WMO: 350675

Lat: 50.037N

Lon: 82.494E

Elev: 939

StdP: 14.20

Time Zone: 6.00 (E06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.9	-33.2	-45.4	0.4	-38.9	-39.6	0.6	-33.7	21.2	28.0	19.5	26.2	1.2	10	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.8	91.4	65.7	88.0	64.7	84.6	64.1	69.3	83.1	67.7	81.7	66.5	80.5	8.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	94.6	74.1	62.9	88.8	72.4	61.2	83.5	71.0	33.8	82.9	32.5	81.6	31.5	80.2	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.5	16.2	DB	-40.8	96.0	6.7	3.2	-45.6	98.3	-49.5	100.1	-53.2	101.9	-58.1	104.2	(4)
(5)			WB	-41.0	73.6	6.6	2.6	-45.7	75.4	-49.5	77.0	-53.2	78.4	-58.0	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	0.1	1.9	22.2	45.4	55.3	66.4	68.4	65.4	54.4	42.1	24.1	8.6
	DBStd	27.03	17.44	16.30	13.61	9.94	8.18	6.62	4.98	5.78	8.19	8.69	14.47	16.40
	HDD50	6367	1548	1348	861	192	41	1	0	46	269	778	1283	
	HDD65	10178	2013	1768	1326	587	315	64	24	67	329	711	1226	1748
	CDD50	2001	0	0	0	55	205	493	570	476	177	23	2	0
	CDD65	337	0	0	0	0	14	106	129	78	10	0	0	0
	CDH74	5557	0	0	0	56	401	1560	1711	1405	403	21	0	0
	CDH80	2050	0	0	0	5	102	596	655	552	139	1	0	0
	WSAvg	5.9	4.4	4.2	5.4	6.7	8.1	6.3	5.9	5.6	5.3	6.0	6.7	5.5
	PrecAvg	18.6	1.0	1.0	1.1	1.2	2.0	1.7	2.0	1.8	1.4	1.7	1.7	1.6
(15) Precipitation	PrecMax	32.6	3.3	3.2	3.3	3.5	4.2	4.9	8.4	6.4	3.9	4.5	3.3	3.5
	PrecMin	12.7	0.3	0.0	0.2	0.1	0.0	0.5	0.2	0.2	0.0	0.5	0.1	0.2
	PrecStd	4.4	0.6	0.7	0.8	0.7	1.1	1.2	1.5	1.2	0.8	0.9	0.7	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	38.8	57.0	80.5	88.0	93.5	95.2	95.2	90.0	77.4	60.4	41.4
		MCWB	32.2	33.6	44.4	54.7	62.0	65.3	67.1	65.0	60.8	54.2	49.0	34.1
	2%	DB	33.8	35.5	48.0	75.0	82.3	89.8	89.9	91.1	83.8	70.1	51.6	37.5
		MCWB	30.4	31.3	41.1	54.0	59.1	65.4	66.9	65.1	58.5	51.9	42.4	33.2
	5%	DB	30.2	31.6	43.0	70.0	78.5	86.3	86.4	86.3	78.4	62.9	46.1	34.1
		MCWB	27.3	28.9	37.8	51.9	57.5	64.6	66.5	64.2	57.5	49.0	40.3	30.6
	10%	DB	26.4	26.6	39.4	64.6	73.8	82.8	83.8	82.4	71.9	57.4	41.0	30.3
		MCWB	24.0	24.4	35.4	50.2	55.9	64.2	65.7	63.1	55.2	47.0	37.2	27.8
	0.4%	WB	33.7	34.5	46.3	58.0	65.7	70.6	72.6	70.8	63.8	57.0	50.1	35.5
		MCDB	36.6	37.2	54.0	73.7	84.1	83.7	85.9	84.7	82.9	70.9	57.4	38.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	30.1	31.4	41.3	55.6	61.8	68.8	69.7	67.9	60.5	52.8	43.7	33.2
		MCDB	34.0	34.7	46.3	69.9	74.8	82.5	83.5	82.7	77.7	68.0	49.3	36.9
	5%	WB	26.8	28.1	38.2	53.4	59.5	67.3	68.0	66.3	58.4	50.1	40.2	30.9
		MCDB	29.7	31.1	42.5	66.8	73.4	81.1	81.4	81.2	74.2	62.1	44.9	34.2
	10%	WB	23.8	24.3	35.8	50.9	57.5	65.6	66.8	64.6	56.5	47.7	37.3	27.5
		MCDB	26.1	26.2	38.9	63.4	70.9	79.5	80.1	78.4	70.2	55.8	40.6	30.4
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	25.2	21.5	26.0	28.7	27.7	26.8	30.0	29.5	22.6	16.8	19.0
		MCDBR	20.9	23.2	22.4	37.6	35.9	34.3	33.9	38.4	40.0	36.2	21.8	17.3
		MCWBR	18.2	20.1	15.9	19.1	17.4	14.4	14.9	15.9	18.2	20.3	14.4	13.9
	5% WB	MCDBR	18.9	22.6	20.4	32.4	28.7	28.6	28.5	31.5	34.4	32.2	19.7	16.8
		MCWBR	16.8	20.2	14.9	18.1	14.8	13.1	13.6	14.6	16.9	18.7	13.8	13.9
(40) Clear-Sky Solar Irradiance	taub	0.304	0.342	0.395	0.433	0.420	0.419	0.440	0.407	0.367	0.368	0.343	0.311	
	taud	2.022	1.949	1.933	2.057	2.155	2.235	2.185	2.257	2.314	2.202	2.076	1.983	
	Ebn at Noon	218	239	249	256	266	267	258	262	260	232	200	191	
	Edh at Noon	28	40	49	48	45	42	44	39	33	31	27	25	
(44) All-Sky Solar Radiation	RadAvg	480	835	1255	1486	1878	2031	1875	1742	1282	698	401	346	
	RadStd	31	36	110	140	146	167	149	117	116	90	43	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page